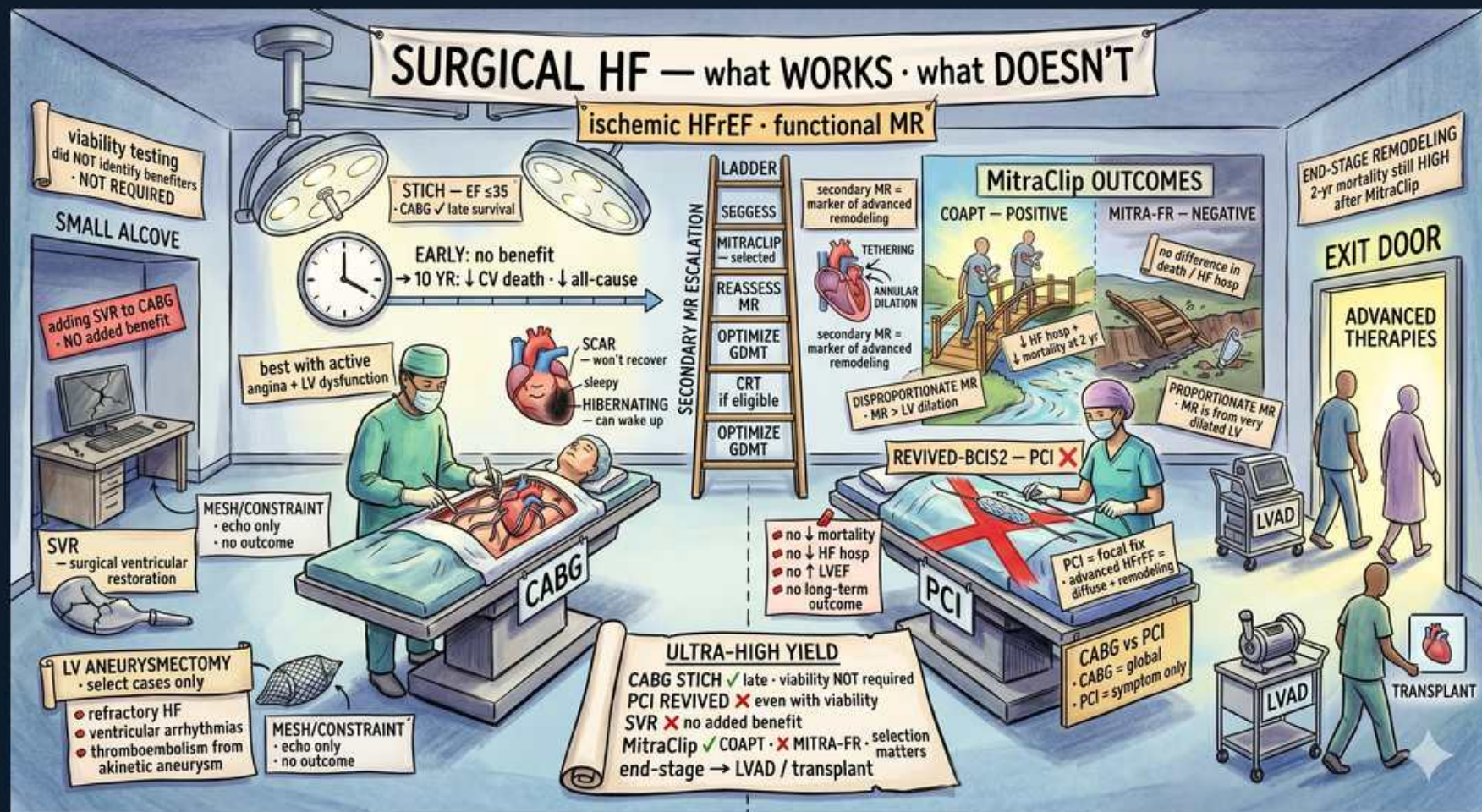


Heart Failure — Surgical & Advanced Therapies



1. Surgical HF — what works / doesn't

WHAT YOU SEE

Title: ischemic HFrEF, functional MR

WHAT IT ENCODES

Surgical/advanced HF therapy turns on two questions: (1) ischemic HFrEF → does revascularization (CABG > PCI) help? and (2) functional/secondary MR → does MitraClip help? The board theme is patient selection and trial evidence (STICH, REVIVED, COAPT vs MITRA-FR), not surgical technique. End-stage refractory disease escalates to LVAD and transplant.

BOARD TRAP

WRONG: 'Any HFrEF patient benefits from surgery.' Benefit is confined to selected subgroups (ischemic HFrEF for CABG; disproportionate secondary MR for MitraClip); offering surgery to all worsens early outcomes.

2. CABG — STICH, EF ≤35

WHAT YOU SEE

A surgeon at a lit operating table labelled 'CABG' with 'STICH — EF ≤35, CABG > late survival' — the bypass operation that earns a survival win, but only late.

WHAT IT ENCODES

STICH trial: in ischemic cardiomyopathy with EF ≤35%, CABG + GDMT reduced all-cause death (and CV death/hospitalization) versus GDMT alone — but only on long-term follow-up (STICHES, 10 years). Benefit is greatest with multivessel disease, active angina, and viable myocardium. Surgical revascularization restores flow to hibernating territory and prevents future infarction/arrhythmia.

BOARD TRAP

WRONG: 'CABG gives an early survival benefit in HFrEF.' STICH showed NO early benefit (early surgical mortality offsets it); the survival advantage only emerged at ~10 years (STICHES).

3. Early: no early benefit

WHAT YOU SEE

Clock: early no benefit, 10yr benefit

WHAT IT ENCODES

Timeline pearl from STICH: the primary endpoint (all-cause mortality) was neutral at the original ~5-year analysis because of upfront perioperative risk, then turned significantly positive by 10 years (STICHES extension). So CABG for ischemic HFrEF is a long-game intervention — counsel patients that survival benefit is delayed, not immediate.

BOARD TRAP

WRONG: 'A neutral 5-year result means CABG doesn't help HFrEF.' Extended follow-up (10 yr) showed a clear mortality benefit — the early curve crossing is expected with surgery.

4. Viability testing NOT required

WHAT YOU SEE

Small alcove: viability testing not required

WHAT IT ENCODES

STICH viability substudy (and STICHES): myocardial viability testing (SPECT/dobutamine echo/PET) did NOT identify which patients gained a survival benefit from CABG — patients benefited regardless of viability status. So routine viability imaging is NOT required to decide on CABG in ischemic HFrEF; the decision rests on coronary anatomy, symptoms, and surgical candidacy.

BOARD TRAP

WRONG: 'You must prove viable myocardium before offering CABG.' The STICH viability substudy found viability did not predict the mortality benefit — it is not a prerequisite for revascularization.

5. Hibernating myocardium wakes up

WHAT YOU SEE

Scar vs hibernating heart, can wake up

WHAT IT ENCODES

Hibernating myocardium is viable but chronically hypoperfused tissue that down-regulates contraction to survive; it can RECOVER function after revascularization, improving regional and global EF. Scar (transmural infarct) is dead and will not recover. The concept rationalizes revascularization, even though STICH showed the benefit isn't reliably predicted by viability testing.

BOARD TRAP

WRONG: 'Akinetic/scarred segments will improve with revascularization.' Only viable/hibernating myocardium recovers; non-viable scar does not — but you still don't need to prove this before CABG (per STICH).

6. PCI — not proven

WHAT YOU SEE

Big red X over PCI, REVIVED-BCIS2

WHAT IT ENCODES

REVIVED-BCIS2: in stable severe ischemic LV dysfunction (EF ≤35%) with extensive viable myocardium, PCI + GDMT did NOT reduce death or HF hospitalization versus GDMT alone, and did not improve EF or quality of life at follow-up. PCI treats focal lesions but does not deliver the complete/global revascularization (and bypass of future culprit lesions) that CABG provides.

BOARD TRAP

WRONG: 'PCI is equivalent to CABG for ischemic HFrEF.' REVIVED-BCIS2 was negative for PCI; the durable survival evidence in ischemic HFrEF belongs to CABG (STICH), not PCI.

7. MitraClip — COAPT positive

WHAT YOU SEE

MitraClip outcomes, COAPT positive

WHAT IT ENCODES

COAPT (transcatheter edge-to-edge repair, MitraClip) was POSITIVE in secondary/functional MR: on top of maximally tolerated GDMT it reduced HF hospitalizations and all-cause mortality at 2 years. The enrolled MR was severe but DISPROPORTIONATE to LV dilation (high regurgitant fraction, smaller LV volumes) — i.e., the MR itself was driving symptoms.

BOARD TRAP

WRONG: 'MitraClip helps any patient with secondary MR.' Benefit (COAPT) is limited to MR that is severe and disproportionate to LV size, on optimized GDMT — not the dilated-LV-with-proportionate-MR group.

8. MITRA-FR negative

WHAT YOU SEE

A MitraClip outcome panel marked 'MITRA-FR — NEGATIVE' with a red X — the X flags the trial where the clip FAILED (dilated LV, proportionate MR).

WHAT IT ENCODES

MITRA-FR was NEGATIVE for the same device: no reduction in death or HF hospitalization. The difference from COAPT is patient selection — MITRA-FR enrolled patients with very dilated LVs and MR PROPORTIONATE to the dilation (less severe regurgitation, larger volumes), where the LV remodeling, not the valve, drives prognosis. This 'proportionate vs disproportionate MR' framework reconciles the two trials.

BOARD TRAP

WRONG: 'COAPT and MITRA-FR conflict, so MitraClip evidence is unreliable.' They differ by design/selection: disproportionate MR + smaller LV (COAPT) benefits; proportionate MR + very dilated LV (MITRA-FR) does not.

9. Optimize GDMT first

WHAT YOU SEE

Reassess, optimize GDMT before clip

WHAT IT ENCODES

Secondary MR is a consequence of LV remodeling, so guideline-directed medical therapy (ARNI/ACEi, beta-blocker, MRA, SGLT2i) ± CRT must be maximized FIRST — reverse remodeling often reduces MR severity and the patient may no longer qualify. Reassess MR after optimization; only persistent severe disproportionate MR on full GDMT meets COAPT-type criteria for MitraClip.

BOARD TRAP

WRONG: 'Severe secondary MR is a structural problem that should go straight to the cath lab.' GDMT (and CRT if eligible) can shrink the LV and the MR; intervention is only for MR that persists after true optimization.

10. SVR — no outcome benefit

WHAT YOU SEE

SVR surgical ventricular restoration, no outcome

WHAT IT ENCODES

Surgical ventricular restoration/reconstruction (STICH Hypothesis 2): adding surgical LV volume-reduction to CABG did NOT improve survival or hospitalization versus CABG alone, despite reducing LV volume. So SVR is not routine; reshaping the ventricle does not translate into outcome benefit.

BOARD TRAP

WRONG: 'Surgical ventricular restoration improves survival in ischemic HF by reducing LV volume.' STICH Hypothesis 2 was neutral — SVR added no outcome benefit over CABG alone.

11. LV aneurysmectomy — selected

WHAT YOU SEE

LV aneurysmectomy select cases

WHAT IT ENCODES

LV aneurysmectomy (resection of a true post-MI aneurysm) is reserved for SELECTED indications, not routine remodeling: refractory heart failure from a large dyskinetic aneurysm, intractable ventricular arrhythmias arising at the aneurysm border zone, or recurrent thromboembolism from mural thrombus in an akinetic sac. Outside these niches it has no proven survival benefit.

BOARD TRAP

WRONG: 'LV aneurysmectomy should be done in most post-MI patients to improve EF.' It is only for refractory HF, refractory VT, or thromboembolism from the aneurysm — not a routine EF-improving operation.

12. LVAD

WHAT YOU SEE

An LVAD pump device by the 'EXIT DOOR / advanced therapies' archway — the exit door = end-stage escalation out of medical management to mechanical support.

WHAT IT ENCODES

A left ventricular assist device (continuous-flow pump, e.g., HeartMate 3) is for end-stage (NYHA IIIb–IV, INTERMACS) HFrEF: used as BRIDGE TO TRANSPLANT in candidates or DESTINATION THERAPY in those ineligible for transplant. Modern devices markedly improve survival/QOL but require lifelong anticoagulation; key complications are GI bleeding (acquired von Willebrand + AV malformations), driveline infection, pump thrombosis, and stroke. Patients are characteristically pulseless (continuous flow → faint/absent pulse, MAP by Doppler).

BOARD TRAP

WRONG: 'An LVAD is a last-ditch comfort measure with no survival benefit.' Modern continuous-flow LVADs substantially improve survival and function as bridge-to-transplant or destination therapy in appropriately selected end-stage HFrEF.

13. Transplant

WHAT YOU SEE

A patient figure walking out through the 'EXIT DOOR' carrying a new heart, labelled 'TRANSPLANT' — walking out the door = the definitive end-stage destination beyond LVAD.

WHAT IT ENCODES

Heart transplantation is the definitive therapy for refractory end-stage HF (cardiogenic shock dependence, refractory NYHA IV, peak $\text{VO}_2 < 14$ / < 12 on beta-blocker) in candidates without prohibitive comorbidity. Contraindications include fixed severe pulmonary hypertension (high PVR), active infection/malignancy, irreversible end-organ damage, and poor adherence. Lifelong immunosuppression brings rejection, infection, malignancy (PTLD), and cardiac allograft vasculopathy (diffuse, denervated → silent ischemia).

BOARD TRAP

WRONG: 'Fixed severe pulmonary hypertension is fine for heart transplant.' High fixed PVR is a major contraindication — the naive donor RV fails acutely against it; such patients may need an LVAD to unload/reduce PVR first.
